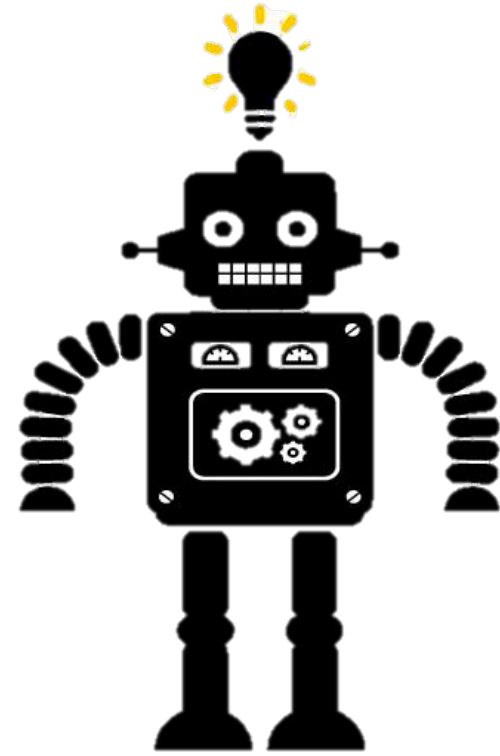


Intro to Deep Learning

Introduction to machine learning

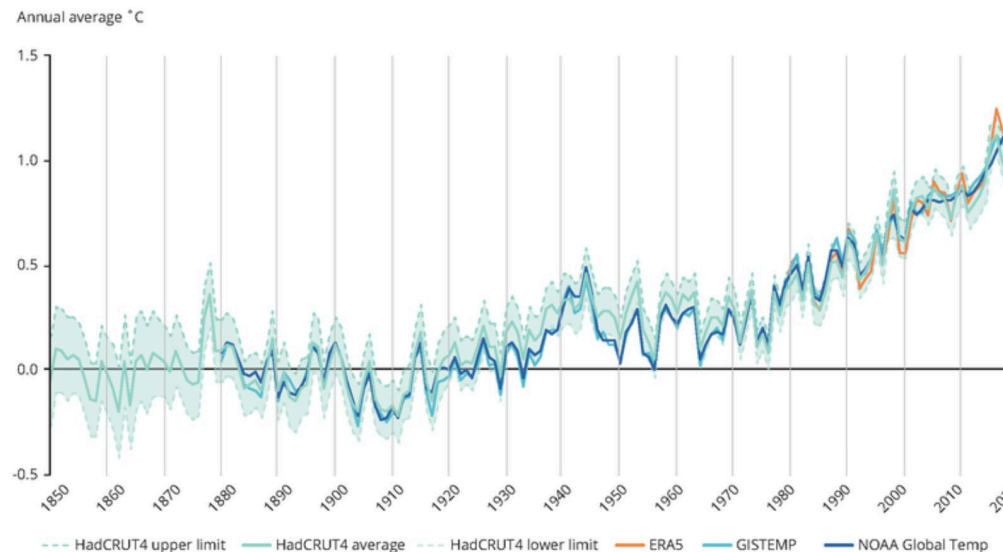


In this lecture

- What is machine learning?
- Why machine learning?
- Examples of machine learning applications
- History of machine learning

What is machine learning?

- Data are our measurements from a system so studying them helps us to make decisions about the system.
- ML is a tool that we use to study and learn from the data.
- Eg., data about the global mean surface shows a trend that enables computers to learn and infer from data that the global temperature is increasing.



Why machine learning works?

- The idea of machine learning works because data is not random – it has structure that can tell us about the underlying behavior of its generating system
- That is why we can use ML and train computers to understand the underlying process & make predictions/decisions for the future

Examples

- Media – Netflix to predict what individual viewers will likely enjoy based on the streaming history and habits of its millions of users.
- Web mining – search engines, Yelp to sort reviewers' photos to groups: menus, food, inside, outside and so on.
- Finance – credit scoring, fraud detection.
- Medicine – medical diagnosis, using electronic medical records, financial data and claims to predict sickness.
- Manufacturing – control, robotics, troubleshooting
- Transportation – self-driving cars.
- Bioinformatics – Motifs, alignments.

Why we need machine learning?

- Humans need to go through lots of training and still are prone to making mistakes
- Human expertise aren't readily available
- Solutions change in time (routing on a computer network)
- Solutions need to be adapted to particular cases
- Humans are unable to explain their expertise
- Human expertise doesn't exist

History of machine learning

- The concept of “computing machinery intelligence” started in 1950s.
- It focused on
 - Human-level intelligence: abstract reasoning
 - Linguistic intelligence
 - Symbol using and knowledge representation
 - Rule-following and logic
- Games are popular because a metric exists to prove one algorithm is working better than another

History of machine learning (Cont.)

- In 1997, a milestone was reached: AI beats best human at chess!
- In 2011, IBM's Watson beats former human winners at the quiz game Jeopardy!
- In 2016, Google DeepMind beats world champions in Go game!



Garry Kasparov makes a move during his fourth game against the IBM Deep Blue chess computer. | Stan Honda/AFP/Getty Images

Why tremendous growth in machine learning?

- With recent technology, data has become cheap!
 - How many surveillance cameras are deployed in the US?
 - What is the size of the World Wide Web?
 - How many youtube videos are uploaded per day?



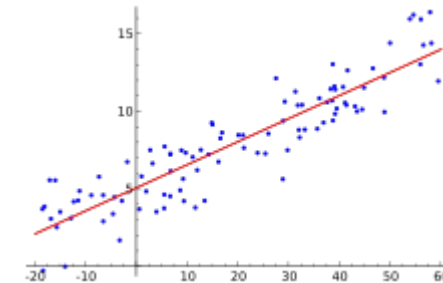
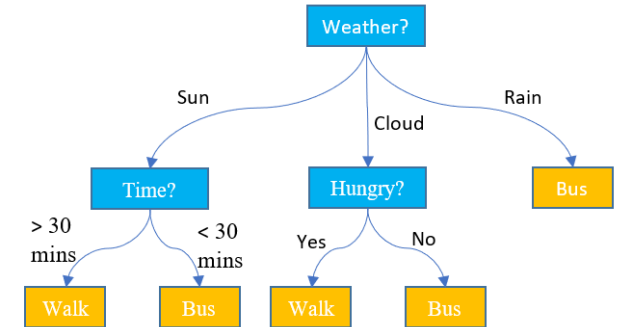
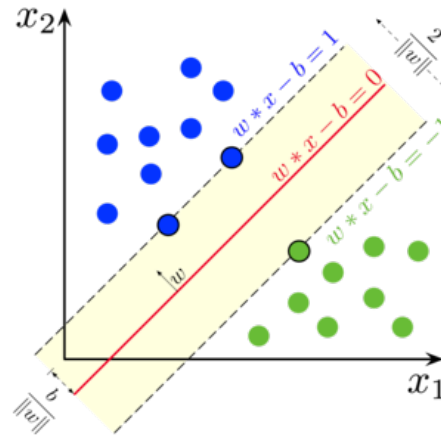
Example applications

- https://www.youtube.com/watch?v=Bg_tJvCA8zw
 - <https://www.youtube.com/watch?v=cltRwEGThvo>
 - <https://www.youtube.com/watch?v=kopoLzvh5jY>
 - <https://www.youtube.com/watch?v=FOlzYBD4vxo>
 - <https://www.youtube.com/watch?v=y1sxsye11xk>
 - <https://www.youtube.com/watch?v=nAMSfmHuMOQ>
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- <https://experiments.withgoogle.com/experiments>
 - <https://teachablemachine.withgoogle.com>
 - <https://experiments.withgoogle.com/body-synth>
 - <https://quickdraw.withgoogle.com/>
 - <https://this-person-does-not-exist.com/>

Different machine learning tools

- Traditional/classic machine learning methods

- Regression
- Logistic regression
- Decision trees
- Support vector machine



- Artificial Neural networks
- Deep learning



History of neural networks

